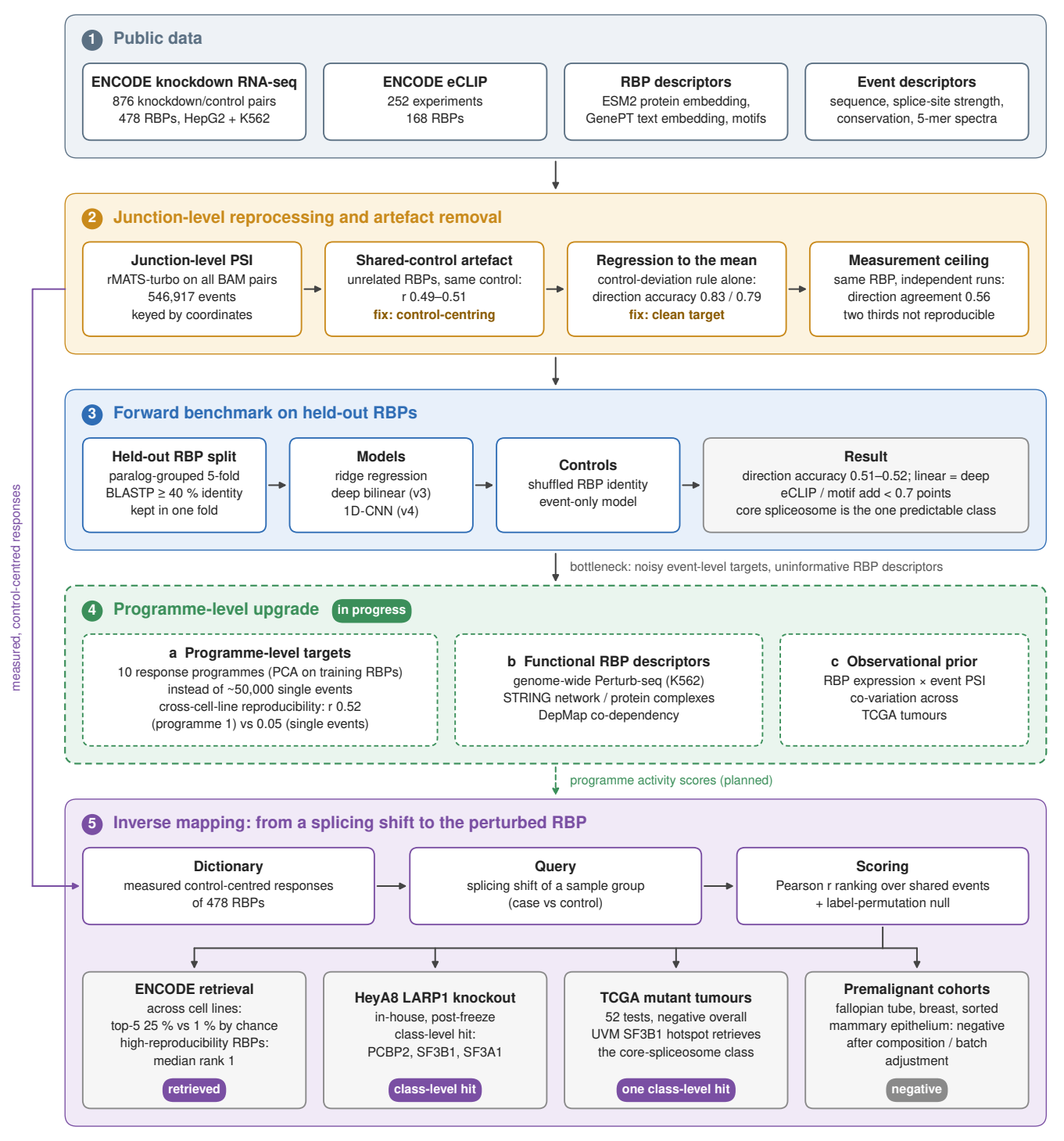
1 Public data
ENCODE knockdown RNA-seq
876 knockdown/control pairs
478 RBPs, HepG2 + K562
ENCODE eCLIP
252 experiments
168 RBPs
RBP descriptors
ESM2 protein embedding,
GenePT text embedding, motifs
Event descriptors
sequence, splice-site strength,
conservation, 5-mer spectra
2 Junction-level reprocessing and artefact removal
Junction-level PSI
rMATS-turbo on all BAM pairs
546,917 events
keyed by coordinates
Shared-control artefact
unrelated RBPs, same control:
r 0.49–0.51
fix: control-centring
Regression to the mean
control-deviation rule alone:
direction accuracy 0.83 / 0.79
fix: clean target
Measurement ceiling
same RBP, independent runs:
direction agreement 0.56
two thirds not reproducible
3 Forward benchmark on held-out RBPs
Held-out RBP split
paralog-grouped 5-fold
BLASTP ≥ 40 % identity
kept in one fold
Models
ridge regression
deep bilinear (v3)
1D-CNN (v4)
Controls
shuffled RBP identity
event-only model
Result
direction accuracy 0.51–0.52; linear = deep
eCLIP / motif add < 0.7 points
core spliceosome is the one predictable class
bottleneck: noisy event-level targets, uninformative RBP descriptors
4 Programme-level upgrade
in progress
a Programme-level targets
10 response programmes (PCA on training RBPs)
instead of ~50,000 single events
cross-cell-line reproducibility: r 0.52
(programme 1) vs 0.05 (single events)
b Functional RBP descriptors
genome-wide Perturb-seq (K562)
STRING network / protein complexes
DepMap co-dependency
c Observational prior
RBP expression × event PSI
co-variation across
TCGA tumours
programme activity scores (planned)
5 Inverse mapping: from a splicing shift to the perturbed RBP
Dictionary
measured control-centred responses
of 478 RBPs
Query
splicing shift of a sample group
(case vs control)
Scoring
Pearson r ranking over shared events
+ label-permutation null
ENCODE retrieval
across cell lines:
top-5 25 % vs 1 % by chance
high-reproducibility RBPs:
median rank 1
retrieved
HepA8 LARP1 knockout
in-house, post-freeze
class-level hit:
PCBP2, SF3B1, SF3A1
class-level hit
TCGA mutant tumours
52 tests, negative overall
UVM SF3B1 hotspot retrieves
the core-spliceosome class
one class-level hit
Premalignant cohorts
fallopian tube, breast, sorted
mammary epithelium: negative
after composition / batch
adjustment
negative
measured, control-centred responses